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SEMI E40.1-1218 (Reapproved 0324)
SPECIFICATION FOR SECS-II PROTOCOL FOR PROCESSING MANAGEMENT

This Standard was technically approved by the Information & Control Global Technical Committee. This edition was approved for publication by the global Audits and Reviews Subcommittee on December 14, 2023. Available at www.semiviews.org and www.semi.org in March 2024. Originally published 1995; previously published December 2018.

1 Purpose

1.1 This Standard maps the services and data of SEMI E40 to SECS-II streams and functions and data definitions.

2 Scope

2.1 This is the standard way to implement the Processing Management, which provides remote control of wafer processing, using the SECS-II message protocol.

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3 Referenced Standards and Documents

3.1 *SEMI Standards and Safety Guidelines*

SEMI E5 — Specification for SEMI Equipment Communications Standard 2 Message Content (SECS-II)

SEMI E30 — Specification for the Generic Model for Communications and Control of Manufacturing Equipment (GEM)

SEMI E39 — Specification for Object Services: Concepts, Behavior, and Services

SEMI E39.1 — Specification for SECS-II Protocol for Object Services (OSS)

SEMI E40 — Specification for Processing Management

NOTICE: Unless otherwise indicated, all documents cited shall be the latest published versions.

4 Terminology

4.1 None.

5 Mapping of Processing Services

Table 1 Processing Management Messages Mapping

Service Message Name	*Stream, Function*	*SECS-II Name*
PRJobCreate request	S16F3,F4	Process Job Create Request/Acknowledge
PrJobCommand request	S16F5,F6	Process Job Command Request/Acknowledge
PRJobAlert notify	If E30 style events: S6F11,F12 If E40 style alerts: S16F7,F8	If E30 style events: Event Report Send/Acknowledge If E40 style alerts: Process Job Alert Notify/Confirm
PRJobEvent notify	If E30 style events: S6F11,F12 If E40 style events: S16F9,F10	If E30 style events: Event Report Send/Acknowledge If E40 style events: Process Job Event Notify/Confirm
PRSetMtrlOrder request	S16F29,30	Process Job Set Material Order
PRJobCreateEnh	S16F11/F12	PRJobCreateEnh

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Service Message Name	*Stream, Function*	*SECS-II Name*
PRJobMultiCreate	S16F15/F16	PRJobMultiCreate
PRJobDequeue	S16F17/F18	PRJobDequeue
PRGetAllJobs	S16F19/F20	PRGetAllJobs
PRGetSpace	S16F21/F22	PRGetSpace
PRJobSetRecipeVariable	S16F23/F24	PRJobSetRecipeVariable
PRJobSetStartMethod	S16F25/F26	PRJobSetStartMethod

6 Mapping of Processing Parameter

Table 2 Data Item Mapping

Service Parameter	*SECS-II Data Item*
PrJobID	PRJOBID
PRMtlType	MF
PRMtlName	MID
PRAck	ACKA
PRRecipeMethod	PRRECIPEMETHOD
RecID	RCPSPEC or PPID; only in ASCII (20) format[#1]
RecipeVarName	RCPPARNM
RecipeVarValue	RCPPARVAL
PRProcessStart	PRPROCESSSTART
PRCmdName	PRCMDNAME
PRJobMilestone	If E30 style events: CEID If E40 style alerts: PRJOBMILESTONE
PRJobState	PRSTATE
Timestamp	TIMESTAMP
PREventID	If E30 style events: CEID If E40 style events: PREVENTID
CmdParmName	CPNAME
CmdParmVal	CPVAL
PRMtrlOrder	PRMTRLORDER
ErrorCode	ERRCODE
ErrorText	ERRTEXT
PREventData	V (SV, ECV, DVVAL)
PRJobSpace	PRJOBSPACE
PRPauseEvent	PRPAUSEEVENT

#1 If an implementer uses PPID for RecID, the format of PPID shall be ASCII because RecID is defined as the text in SEMI E40.

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Table 3 Service Parameter Mapping

Service Parameter	*SECS-II Format*
CmdParameter	List of command parameter names and values. L,n 1. L,2 1. <CPNAME$_1$> 2. <CPVAL$_1$> … n. L,2 1. <CPNAME$_n$> 2. <CPVAL$_n$>
PRJobIDList	List of process job identifiers. L,n 1. <PRJOBID$_1$> … n. <PRJOBID$_n$>
PRJobList	List of process job identifiers and their states. L,n 1. L,2 1. <PRJOBID$_1$> 2. <PRSTATE$_1$> … n. L,2 1. <PRJOBID$_n$> 2. <PRSTATE$_n$>
PRJobMultiSpec	List of process job specifications L,n 1. L,6 1. <PRJOBID$_1$> 2. <MF$_1$> 3. <PRMtlNameList$_1$> 4. <PRRecipe$_1$> 5. <PRPROCESSSTART$_1$> 6. <PRPAUSEEVENT$_1$> … n. L,6 1. <PRJOBID$_n$> 2. <MF$_n$> 3. <PRMtlNameList$_n$> 4. <PRRecipe$_n$> 5. <PRPROCESSSTART$_n$> 6. <PRPAUSEEVENT$_n$>

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Service Parameter	*SECS-II Format*
PRRecipe	L,3 1. <PRRECIPEMETHOD> 2. <RCPSPEC> or <PPID>; In the case of <PPID>, only in ASCII (20) format[#1] 3. L,n n = # of Recipe Variables 1. <RecipeVariable$_1$> … n. <RecipeVariable$_n$>
PRStatus	L,2 1. <ACKA> 2. L,n 1. <Status$_1$> … n. <Status$_n$>
RecipeVariable	L,2 1. <RCPPARNM> 2. <RCPPARVAL>
Status	L,2 1. <ERRCODE> 2. <ERRTEXT>

#1 If an implementer uses PPID for PRRecipe, the format of PPID shall be ASCII because RecID is defined as the text in SEMI E40.

7 Variable Data Item Mapping

7.1 This section shows the specific SECS-II data classes, and formats needed for SECS-II implementations of SEMI E40 variable data. According to SEMI E40 § 11, all ProcessJob object attributes are to be available as variables for Process Job state transition events. These variables will be of SEMI E5 data item DVVAL.

8 Implementation Details

8.1 *Use of Object Services* — Several capabilities of the Processing Management Services are accessed through the Object Services Standard. When a Process Job has been created, (PRJOBID is valid), then its attributes can be read and written using the Object Services GetAttr and SetAttr messages.

8.1.1 SEMI E39 Object Services shall be used for access to ProcessJob attributes. The GetAttr service may be used for all ProcessJob attributes and the SetAttr service may be used on only those attributes whose Access is set to RW.

8.2 *Multi-Block Messages* — Processing Management Services is protocol independent and therefore, makes no mention of SECS-II multi-block access and grant messages. When these Service use the SECS-II protocol, then S16F3,F5 shall be preceded by an S16F1/S16F2 access request/grant message exchange when the message will be multi-block.

9 SECS-II Attribute Definitions

9.1 *Process Job Object SECS-II Attributes Definitions* — The following are the SECS-II structure definitions for the SEMI E40 ProcessJob object.

Table 4 Process Job SECS-II Attribute Definitions

Attribute Name	*Attribute Data Form: SECS-II Structure*
‘ObjID’	<PRJOBID> (Conforms to the restrictions of ObjID as specified in SEMI E39.1, § 6.)
‘ObjType’	‘ProcessJob’

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Attribute Name	*Attribute Data Form: SECS-II Structure*
'PauseEvent'	L,n n=number of collection events 1. <CEID$_1$> … n. <CEID$_n$> CEID restricted to format U()
'PRJobState'	<PRSTATE> PRJobState PRSTATE enumerated as follows: 51 (U1) Enumerations: 0 – QUEUED/POOLED 1 – SETTING UP 2 – WAITING FOR START 3 – PROCESSING 4 – PROCESS COMPLETE 5 – (Reserved) 6 – PAUSING 7 – PAUSED 8 – STOPPING 9 – ABORTING 10 – STOPPED 11 – ABORTED
'PRMtlNameList'	When MF = 13 (0x0d) (carriers) L,n n=number of carriers 1. L,2 1. <CARRIERID$_1$> 2. L,j j=number of slots 1. <SLOTID$_1$> : j. <SLOTID$_j$> : n. L,2 1. <CARRIERID$_n$> 2. L,k k=number of slots 1. <SLOTID$_1$> : k. <SLOTID$_k$> When MF = 14 (0x0e) (substrate) L,n n=number of material (substrate) 1. <MID$_1$> substrate ID … n. <MID$_n$> MID restricted to format A
'PRMtlType'	<MF> MF restricted to format B **MF Value** **Description** 13 (0x0d) carriers (e.g., FOUP, SMIF pod, cassette) 14 (0x0e) substrate (e.g., wafer, mask, flat panel) other not valid for SEMI E40 material
'PRProcessStart'	<PRPROCESSSTART>
'PRRecipeMethod'	<PRRECIPEMETHOD>
'RecID'	<RCPSPEC> or <PPID>; only in ASCII (20) format[#1]

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Attribute Name	*Attribute Data Form: SECS-II Structure*
'RecVariableList'	L,n n=number of recipe variables 1. L,2 1. <RCPPARNM$_1$> 2. <RCPPARVAL$_1$> … n. L,2 1. <RCPPARNM$_n$> 2. <RCPPARVAL$_n$>

#1 If an implementer uses PPID for RecID, the format of PPID shall be ASCII because RecID is defined as the text in SEMI E40.

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